he did not, at that early stage of understanding, incorporate special relativity.4

But physicists soon realized that special relativity was central to a proper quantum-mechanical framework. This is because the microscopic frenzy requires that we recognize that energy can manifest itself in a huge variety of ways—a notion that comes from the special relativistic declaration E = mc

2. By ignoring special relativity, Schrödinger's approach ignored the malleability of matter, energy, and motion.

Physicists focused their initial pathbreaking efforts to merge special relativity with quantum concepts on the electromagnetic force and its interactions with matter. Through a series of inspirational developments, they created quantum electrodynamics. This is an example of what has come to be called a relativistic quantum field theory, or a quantum field theory, for short. It's quantum because all of the probabilistic and uncertainty issues are incorporated from the outset; it's a field theory because it merges the quantum principles into the previous classical notion of a force field—in this case, Maxwell's electromagnetic field. And finally, it's relativistic because special relativity is also incorporated from the outset. (If you'd like a visual metaphor for a quantum field, you can pretty much invoke the image of a classical field—say, as an ocean of invisible field lines permeating space—but you should refine this image in two ways. First, you should envision a quantum field as composed of particulate ingredients, such as photons for the electromagnetic field. Second, you should imagine energy, in the form of particles' masses and their motion, endlessly shifting back and forth from one quantum field to another as they continually vibrate through space and time.)

Quantum electrodynamics is arguably the most precise theory of natural phenomena ever advanced. An illustration of its precision can be found in the work of Toichiro Kinoshita, a particle physicist from Cornell University, who has, over the last 30 years, painstakingly used quantum electrodynamics to calculate certain detailed properties of electrons. Kinoshita's calculations fill thousands of pages and have ultimately required the most powerful computers in the world to complete. But the effort has been well worth it: the calculations yield predictions about electrons that have been experimentally verified to